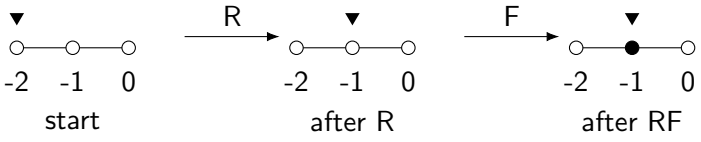
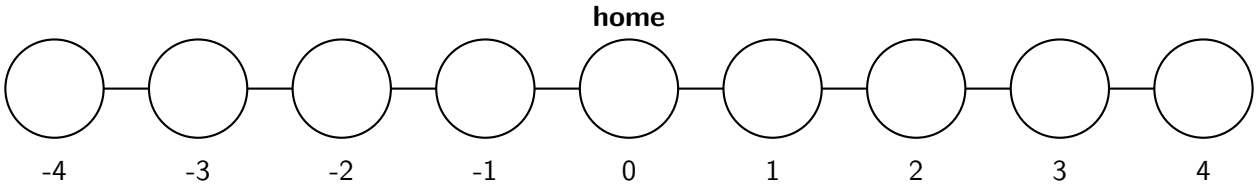
One move is **L**: walk left one place, **R**: walk right one place, or **F**: flip the lamp under the walker. ○ is off, ● is on, and ▼ is the walker. Every target starts with every lamp off and the walker at home, 0.

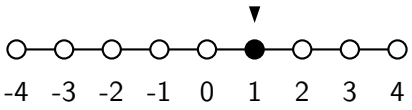
One partner moves the walker. The other flips the lamp, only after pointing to the walker's place. Write one letter per move. Swap jobs between targets.



Problem 1: Reach each target in as few moves as you can. Reset before each attempt.



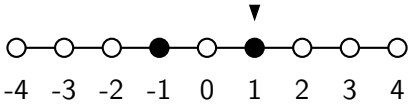
A



Move word: _____

Fewest moves: _____

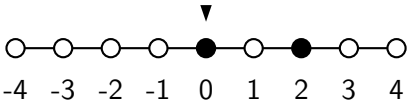
B



Move word: _____

Fewest moves: _____

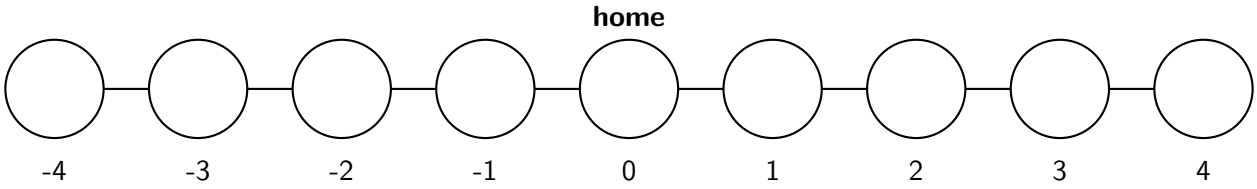
C



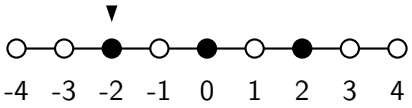
Move word: _____

Fewest moves: _____

Problem 2: Find the fewest moves to each target.



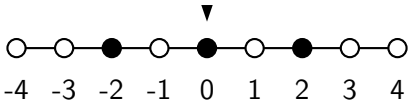
A



Move word: _____

Fewest moves: _____

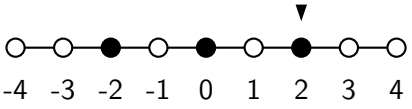
B



Move word: _____

Fewest moves: _____

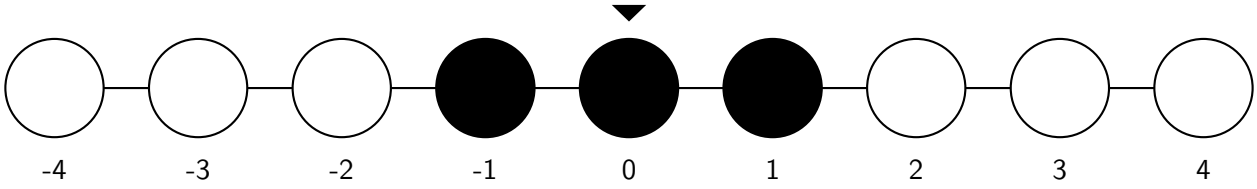
C



Move word: _____

Fewest moves: _____

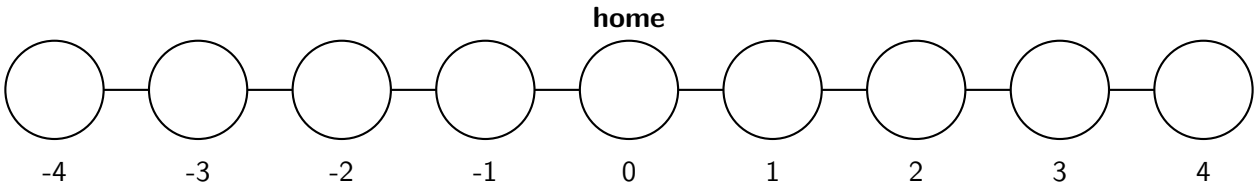
Problem 3: Reach this target in the fewest moves. Find a way to convince your partner that fewer moves cannot work.



Move word: _____

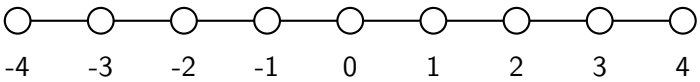
Fewest moves: _____

Workspace



Problem 4: Make one move from the target in Problem 3. Draw the new lamps and walker below. Find the fewest moves from **all lamps off, walker at 0** to this new target. Reset to the Problem 3 target before trying L, R, and F.

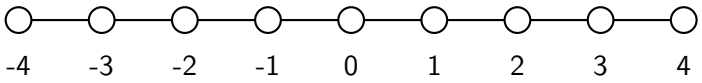
L



Move word: _____

Fewest moves: _____

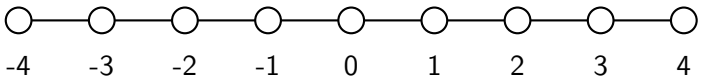
R



Move word: _____

Fewest moves: _____

F

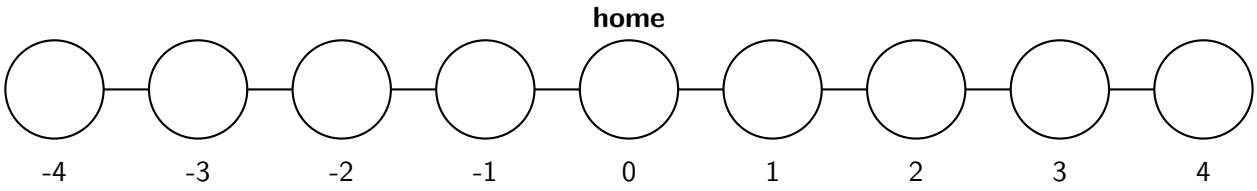


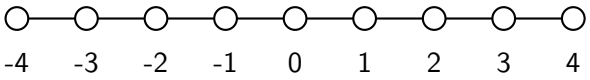
Move word: _____

Fewest moves: _____

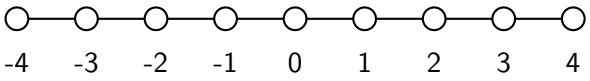
Problem 5: Does every target have at least one next move, L, R or F, that makes it harder to reach from all-off at 0? Use your results to settle this question.

Problem 6: Find other targets where L, R and F each make the target easier to reach from all-off at 0. Which targets are like this?

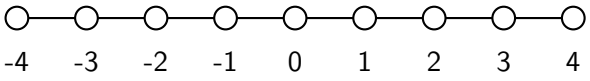




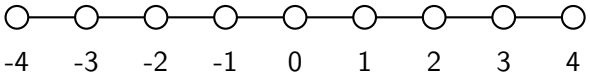
Fewest _____ After L _____ R _____ F _____



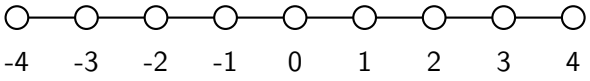
Fewest _____ After L _____ R _____ F _____



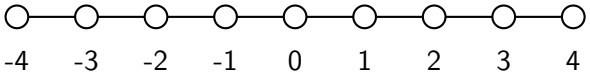
Fewest _____ After L _____ R _____ F _____



Fewest _____ After L _____ R _____ F _____



Fewest _____ After L _____ R _____ F _____



Fewest _____ After L _____ R _____ F _____